📱 Nexora Mobile — UI/UX Product Design Specification & Handover Document

Document Version: 2.0 (Production-Ready Architecture)

Project Location: `A:\Projects\Nexora\mobile\nexora_mobile`

Target Platforms: iOS & Android (Mobile First, Material 3 + iOS Human Interface Guidelines)

Target Audience: Individuals and couples seeking a private, secure, all-in-one personal operating system with AI intelligence, shared planning, encrypted knowledge storage, and news curation.

Deliverables Required From Designer: Figma Component Library, Auto-layout Design Kits, Dark & Light Mode Prototypes, Interactive User Journeys.

---

1. Product Vision & Design Identity

1.1 Core Value Proposition

Nexora is an intelligent personal operating system and private couple space. It eliminates app fragmentation by integrating:

1. Personal Productivity: Dynamic note-taking with pinning, tags, and multi-line editor.

2. Private Couple Space: Dual-user pairing via cryptographic 8-character invite codes, shared notes, and memory milestones.

3. Dual-Layer Calendar: Private individual events alongside synced couple events.

4. Private Couple Chat: Direct, encrypted communication channel between partners.

5. Knowledge Vault: Biometric-ready encrypted storage for PINs, credentials, identity cards, and sensitive documents.

6. AI Assistant & Relationship Coach: Context-aware LLM assistance plus relationship and communication coaching.

7. Curated Intelligence Feed: Tagged news covering AI, Tech, Science, Business, and Lifestyle with Top Stories filtering.

8. Universal Search Index: Single search query retrieving relevant notes, events, vault items, and news simultaneously.

1.2 Visual Tone & Aesthetic

- **Look & Feel:** High-end, futuristic yet warm. A blend of **Notion** (clean layout and typography) + **Arc Browser** (playful modern gradients) + **Between / Paired** (intimate, warm couple aesthetics).
- **Keywords:** Sleek, Private, Intelligent, Fluid, Reassuring.

---

2. Global Design System Foundations

2.1 Color Palette & Semantic Tokens

Token Name	Light Mode Value	Dark Mode Value	Semantic Role
`Brand Primary (Deep Purple)`	`#6C5CE7`	`#8274ED`	App brand identity, AI actions, primary CTA
`Couple Accent (Warm Pink)`	`#FD79A8`	`#FF8FB8`	Couple Space, shared events, anniversary badges
`Vault Accent (Amber Gold)`	`#E17055`	`#F39C12`	Security badge, credentials, locked records
`Intelligence (Electric Blue)`	`#0984E3`	`#48DBFB`	News, external links, global search highlights
`Success (Emerald Green)`	`#00B894`	`#55EFC4`	Completed tasks, connected partner status
`Surface Canvas`	`#F8F9FE`	`#0D0E15`	App screen background
`Card / Container Surface`	`#FFFFFF`	`#161822`	Elevated modules, bento boxes, modals
`Stroke / Divider`	`#ECEFF8`	`#222538`	Card borders, table dividers
`Text Primary`	`#1E202B`	`#F5F6FA`	Primary headlines, bold figures
`Text Secondary`	`#718096`	`#A0AEC0`	Subtitles, timestamps, metadata

2.2 Typography Hierarchy

- **App Headlines (Screen Titles):** *Plus Jakarta Sans / Inter*, Bold, `24px - 30px`.
- **Section Headers (Bento Card Titles):** Semi-Bold, `16px - 18px`.
- **Body Text (Notes, Chat, Vault payloads):** Regular / Medium, `14px - 15px`.
- **Monospace / Secure Data (Invite codes, PINs):** *JetBrains Mono*, Bold, `18px - 24px` with `3px` letter-spacing.
- **Badges & Chips:** Semi-Bold, `11px - 12px`, uppercase or title case.

2.3 Component Standards

- **Border Radius:**
- Small (Buttons, Input fields, Chips): `12px`.
- Medium (List items, Chat bubbles): `16px`.
- Large (Bento cards, Bottom sheets): `24px`.
- **Touch Targets:** Minimum `48dp × 48dp` for all buttons and interactive icons.
- **Feedback Modals:** Bottom sheets with grab handles for quick creation; centered blur-backed dialogs for security authorizations.

---

3. Production Information Architecture (IA)

```
graph TD
Splash["Launch & Auth Check"] --> Login["Screen 1: Login / Register"]
Login --> MainShell["Production Hub: Home Dashboard (Bento Grid)"]

MainShell --> SecNotes["Screen 2: Notes Hub & Markdown Editor"]
MainShell --> SecCouple["Screen 3: Couple Space (Pairing / Connected)"]
MainShell --> SecCal["Screen 4: Dual Calendar (Private & Shared)"]
MainShell --> SecChat["Screen 5: Couple Real-time Chat"]
MainShell --> SecVault["Screen 6: Knowledge Vault (Encrypted Storage)"]
MainShell --> SecAI["Screen 7: AI Assistant & Communication Coach"]
MainShell --> SecNews["Screen 8: Curated Intelligence Feed"]
MainShell --> SecSearch["Screen 9: Global Universal Search"]

SecNotes --> NoteEditor["Screen 2B: Full-Screen Note Editor"]
SecVault --> VaultNew["Modal: New Secret / Document"]
SecCal --> CalNew["Modal: New Event (Private vs Shared)"]
```

4. Screen-by-Screen Detailed UI/UX Specifications

SCREEN 1: Authentication & Onboarding

Files: `lib/screens/login_screen.dart`, `lib/screens/register_screen.dart`

1. Purpose

Secure identity gateway. Authenticates the user and initiates secure token storage (`token_storage.dart`).

2. Layout & Elements

- **Brand Hero Banner:**
- Radiant Nexora logo (Deep Purple with subtle pink glow).
- Headline: "*Welcome to Nexora*".
- Tagline: "*Your private intelligence operating system*".
- **Form Inputs:**
- **Full Name** (Register screen only): Text field with person icon.
- **Email:** Email input field with validation.
- **Password:** Password field with trailing Show/Hide eye icon.
- **Interactive Controls:**
- **Primary CTA:** Full-width pill button: `"Sign In"` / `"Create Account"` .
- **State Toggle Button:** `"Don't have an account? Sign Up"` / `"Already registered? Sign In"` .
- **States & Edge Cases:**
- *Loading:* Primary button transitions to centered circular spinner.
- *Error Feedback:* Floating SnackBar with server message (e.g., "*Invalid email or password*").

SCREEN 2: Home Dashboard (The Executive Bento Hub)

File: `lib/main.dart` (`HomeScreen`)

1. Purpose

The launchpad of the application. Upgrades standard raw vertical buttons to a modern **FinTech/Productivity Bento Grid** that provides glanceable widgets for all 8 modules.

2. Top App Bar

- **Left:** User Greeting (`"Good morning, {FullName}"`) + Avatar with online status indicator.
- **Right Actions:**
- **Search Icon:** Opens Global Search directly.
- **Profile / Logout Icon:** Triggers session logout with confirmation prompt.

3. Bento Grid Modules (Interactive Cards)

1. Couple Space Hero Widget (Top Prominence - Pink Gradient Card):

- *If Unpaired:* Heart icon + "*Connect with your partner*" + "*Generate Code*" CTA.
- *If Paired:* Partner's photo/name + "*Connected since Oct 2026*" + Unread chat pill.

2. Notes Widget (Twin Column Left - Purple Border):

- Icon: Note pad + Count badge (`X Notes`).
- Quick preview of the most recently pinned note title.
- `+ Quick Note` shortcut tap target.

3. Shared Calendar Widget (Twin Column Right - Teal Border):

- Next upcoming event title + Time pill + Shared/Private badge.

4. AI Assistant & Coach Card (Wide Horizontal Card):

- Sparkle AI icon + "*Ask Nexora AI anything or get communication advice*".
- Input chip preview: "*Try: Help me draft a message...*".

5. Knowledge Vault Pill Card (Amber Accent):

- Shield lock icon + "*Knowledge Vault*" + Total items count + Subtitle: "*Encrypted & Protected*".

6. Curated News Mini-Feed:

- Headline ticker showing latest top story from AI/Tech feed.

SCREEN 3: Personal Notes & Markdown Editor

Files: `lib/screens/notes_screen.dart` , `lib/screens/note_editor_screen.dart`

1. Purpose

Organized personal knowledge capture supporting pinned notes, multi-line content, and quick editing.

2. Screen 3A: Notes List (`notes_screen.dart`)

- **Header:** Title `"My Notes"` + Search icon + Refresh button.
- **Floating Action Button:** Circular `+` button in deep purple.
- **List / Grid Structure:**
- Card elements:

- **Pin Badge:** Push-pin icon in top-right (purple if pinned, transparent if unpinned).
- **Title:** Bold (`16px`).
- **Snippet:** 2-line preview with ellipsis.
- **Timestamp:** Formatted local time.
- **Actions:**
- **Tap:** Opens full-screen Note Editor.
- **Pin Toggle:** Pin/Unpin icon button.
- **Delete Action:** Trash icon with confirmation dialog.
- **Edge States:**
- *Empty State:* Friendly illustration + "*No notes yet. Tap + to capture your thoughts.*"
- *Error State:* Error text + `"Retry"` button.

3. Screen 3B: Note Editor Screen (`note_editor_screen.dart`)

- **Top Navigation:**
- Back icon (auto-prompts save on exit if modified).
- Right: Pin/Unpin toggle icon.
- Right: `"Save"` text button or checkmark icon.
- **Form Inputs:**
- Title field: Borderless, large bold font (`22px`), placeholder `"Note Title"` .
- Content field: Borderless, multi-line expanding canvas, placeholder `"Start typing..."` .

SCREEN 4: Couple Space (Partner Pairing & Relationship Hub)

File: `lib/screens/couple_screen.dart`

1. Purpose

A private shared environment between two users. Supports cryptographic invite pairing, shared status, and milestone tracking.

2. State A: Unpaired / Pairing Flow

1. Illustration & Explainer:
- Linked rings / double heart illustration.
- Title: "*Connect with Your Partner*".
- Subtitle: "*Pair with exactly one partner to unlock Couple Mode, shared calendar, shared notes, and memory vault.*"

2. Step 1 Card: "Invite Your Partner":
- Button: `"Generate Invitation Code"` (with loading spinner).
- **Generated Code Container:**
- Prominent high-contrast container with `24px` bold monospace letters (e.g., `AB12CD34`).
- Single-tap **"Copy to Clipboard"** icon.
- Footnote: "*Valid for 48 hours. Single use only.*"

3. Step 2 Card: "Enter Partner's Code":
- Uppercase text field with hint `"e.g. XY98ZW12"` .
- Primary Button: `"Connect Couple"` (Warm Pink background, `#FD79A8`).

3. State B: Connected Couple View

- **Celebration Banner:**
- Heart avatar combining user & partner profile initials.
- Title: "*Connected Couple*".
- Partner Name: `"{Partner Full Name}"`.
- Milestone: `"Connected since: {Date}"` + Days together counter badge.
- Active Status Chip: `"Couple Mode Active"` (Pink gradient with heart).
- **Shared Actions Grid:**
- Shortcut buttons to: `"Couple Chat"`, `"Shared Calendar"`, `"Shared Vault Items"`.

SCREEN 5: Dual Calendar (Private & Shared Schedules)

File: `lib/screens/calendar_screen.dart`

1. Purpose

Unified time management allowing users to track private obligations alongside shared couple events without leaking private schedule details.

2. Layout & Elements

- **AppBar:** Title `"Calendar"` + Month/Week view switcher + Refresh icon.
- **Floating Action Button:** `+` Event icon.
- **Event Card Elements:**
- **Time Block:** Start time to End time (`09:00 AM - 10:30 AM`).
- **Event Title:** Bold headline.
- **Location Badge:** Map pin icon + location name (if provided).
- **Visibility Indicator:**
- *Private Event:* Grey lock icon + `"Private (Only You)"`.
- *Shared Event:* Pink heart icon + `"Shared with Partner"`.
- **Delete Button:** Trash icon with quick delete confirmation.
- **"Add Event" Modal Bottom Sheet:**
- Title input (`"Event Title"`).
- Location input (`"Location (optional)"`).
- Description input (`"Notes / Agenda"`).
- Date & Time pickers (Start & End).
- **Visibility Dropdown / Segmented Switch:**
- Option 1: `🔒 Private (Only You)`
- Option 2: `💖 Shared (With Partner)`
- CTA: `"Save Event"`.

SCREEN 6: Real-time Couple Chat

File: `lib/screens/chat_screen.dart`

1. Purpose

An intimate, clutter-free messaging experience exclusively for the paired couple.

2. Top App Bar

- Partner's Name + Avatar.
- Status indicator (e.g., `"Connected"` with green dot).
- Refresh button.

3. Message Stream (Bubble Layout)

- **Outgoing Messages (Current User):**
- Aligned Right.
- Bubble Color: Deep Purple / Brand gradient (`#6C5CE7`).
- Text: White (`15px`).
- Timestamp: Small muted white at bottom right of bubble.
- **Incoming Messages (Partner):**
- Aligned Left.
- Bubble Color: Warm Soft Pink / Neutral Dark Surface.
- Text: Primary text color.
- Timestamp: Muted timestamp.
- **Empty State:** Illustrated conversation bubble + "*Say something sweet to your partner...*".

4. Bottom Input Bar (Sticky)

- Text input field: Rounded pill (`24px` radius), multi-line expandable, hint `"Message partner..."` .
- Trailing Send Button: Circular button with arrow icon; animates when text is present.

SCREEN 7: Knowledge Vault (Secure Encrypted Storage)

File: `lib/screens/vault_screen.dart`

1. Purpose

A secure repository for storing sensitive information: credentials, passwords, card PINs, identity files, and confidential notes.

2. Layout & Elements

- **AppBar:** Title `"Knowledge Vault"` + Shield icon + Refresh.
- **Search Bar:**
- Top sticky search input: `"Search vault titles or tags..."` .
- **Category / Type Badges (Filter Tabs):**
- `All` | `Documents` | `Secrets / Passwords` | `Cards / IDs` .
- **Vault Item Cards:**
- Leading icon depending on type:
- Document: File text icon.
- Secret/Password: Key/Lock icon.
- Card: Credit card icon.
- Title & Description.

- **Secret Payload Display:**
- Displayed as obscured dots (`••••••••••••`).
- **Eye Toggle Icon Button:** Tapping toggles between obscured and visible plain text.
- **Quick Copy Icon Button:** Copies secret to clipboard with auto-clear warning.
- **Visibility Tag:** `[Private]` vs `[Shared with Partner]` .
- **Tags Chips:** Horizontal scroll of tags (e.g., `#finance` , `#home` , `#passwords`).
- **"Store in Vault" Dialog / Bottom Sheet:**
- Title field.
- Description field (optional).
- Item Type Dropdown:
- `0 = Document / Note`
- `1 = Secret / Password / PIN` (enables secret masking)
- `2 = Card / Identification`
- Secret Payload text area.
- Tags input field (comma separated).
- Visibility selector (`Private` vs `Shared with Partner`).
- Action CTA: `"Save to Vault"` .

SCREEN 8: AI Assistant & Communication Coach

File: `lib/screens/ai_screen.dart`

1. Purpose

A dual-purpose intelligence center featuring a contextual AI chat assistant and an automated relationship communication coach.

2. Top Navigation Tabs

1. Tab 1: "Ask AI" (General Intelligence)

2. Tab 2: "Relationship Coach" (Communication Advice)

3. Tab 1: "Ask AI" Interface

- **Context Input (Optional Accordion Card):**
- Expandable card with label: "*Inject Context (e.g. Schedule, Notes, Background)*".
- **Prompt Input Area:**
- Multi-line text field with placeholder "*Ask anything (e.g. plan our weekend trip, summarize my day)...*".
- Action CTA: `"Ask AI"` (Deep Purple button with sparkle icon).
- **AI Output Card:**
- Markdown-rendered response box.
- Copy response icon button.
- Model attribution tag (e.g., `"Powered by Nexora Intelligence"`).

4. Tab 2: "Relationship Coach" Interface

- **Situation Description Card:**
- Label: "*What is the situation or conflict?*" (e.g., "*We disagreed about holiday planning*").

- **Draft Message Input Card:**
- Label: "*What message do you plan to send?*".
- **Action CTA:**
- Button: `"Get Coach Feedback"` (Warm Pink button).
- **Coach Analysis Card (Output):**
- **Tone Assessment Badge:** (e.g., `Empathetic`, `Defensive`, `Constructive`).
- **Actionable Advice:** Bullet points offering tips to improve clarity and compassion.
- **Suggested Rewrite Option:** One-tap copyable alternative phrasing.

SCREEN 9: Curated Intelligence & News Feed

File: `lib/screens/news_screen.dart`

1. Purpose

Clean, distraction-free news aggregator delivering updates across science, tech, AI, and lifestyle without ad bloat.

2. Top App Bar & Filters

- **AppBar:** Title `"Curated News Feed"`.
- **Right Actions:**
- **Star Toggle Icon:** Filters feed to show **"Top Stories Only"** (star turns gold when active).
- **Refresh Icon:** Pulls latest syndicated articles.
- **Horizontal Category Carousel (Choice Chips):**
- `All` | `Tech` | `AI` | `Business` | `Science` | `World` | `Health` | `Lifestyle`.

3. Article Cards Feed

- **Image / Banner (if available):** Rounded `12px` top banner.
- **Category Pill:** Small upper badge (e.g., `TECH`, `AI`).
- **Title:** Bold headline (`16px - 18px`).
- **Summary Snippet:** 2-3 line brief.
- **Metadata Row:**
- Source name (e.g., "*TechCrunch*", "*Nature*").
- Publication time (e.g., "*2 hours ago*").
- External link button to read full article.

SCREEN 10: Global Universal Search & Intelligence

File: `lib/screens/search_screen.dart`

1. Purpose

Omni-channel search bar that queries across all personal and shared records in a single request.

2. Search Input Header

- Prominent search text field with clear `(X)` icon and magnifying glass.
- Placeholder: "*Search notes, events, vault, news, and reminders...*".

3. Unified Results Stream

- **Result Card Component:**
- **Color-Coded Type Badge:**
- `Note`: Blue icon + badge.
- `Calendar Event`: Green calendar icon + badge.
- `Vault Secret`: Purple shield icon + badge.
- `News`: Teal newspaper icon + badge.
- `Reminder`: Orange clock icon + badge.
- **Result Title & Match Snippet:** Highlighted query matches.
- **Relevancy Score / Timestamp.**
- **Tap Interaction:** Direct deep-link to the target screen (e.g., tapping a note opens the Note Editor).
- **Empty State:**
- Pre-search: Quick suggestion chips (e.g., *"Recent Searches"*, *"Passwords"*, *"Anniversary"*).
- No results found: Magnifying glass illustration + *"No matching records found for '{query}'"*.

---

5. UI/UX Micro-Interactions & State Matrix

User Action	Visual Feedback	Micro-interaction Details
Copy Invite Code	Monospace text flashes purple	Haptic tick; SnackBar: *"Invite code copied to clipboard"*
Reveal Vault Secret	Eye icon toggles open; dots dissolve to plain text	180ms fade-in transition
Partner Connects	Heart icon scales up with pulse animation	Full-screen subtle confetti / celebration micro-modal
Send Message	Bubble pops into bottom of chat list	Instant optimistic UI insert; scroll auto-snaps to bottom
Toggle Top Stories	Star icon turns solid Amber (`#F39C12`)	Feed smoothly cross-fades to filtered list
Pin Note	Pin icon turns deep purple; card moves to top	List item re-orders with smooth layout animation

---

6. Designer Checklist & Deliverables Expected

When handing this specification to your UI/UX designer, request the following Figma deliverables:

- [ ] **Figma Design System / Style Guide:**
- Color Styles (Deep Purple, Couple Pink, Vault Amber, Canvas tokens for Light and Dark modes).
- Typography Hierarchy (Headers, Body, Monospace codes).
- Form Elements (Inputs, Eye toggles, Dropdowns, Choice chips).
- Interactive Component Sets (Buttons, Cards, Chat bubbles, Bottom navigation).
- [ ] **Flow 1 — Authentication:** Login, Registration, Validation states.
- [ ] **Flow 2 — Main Bento Dashboard:** Production Hub with all 8 module cards and user header.
- [ ] **Flow 3 — Couple Experience:** Unpaired Invitation Generator, Code Input, Connected Milestone View, Couple Chat stream.
- [ ] **Flow 4 — Productivity & Schedules:** Notes List, Full-screen Note Editor, Dual-Calendar (Private vs Shared filters).
- [ ] **Flow 5 — Security & Vault:** Knowledge Vault list, Reveal/Hide password states, New item modal.

- [ ] **Flow 6 — AI & Intelligence:** "Ask AI" tab, "Relationship Coach" input/output cards, News feed category chips, Universal Search results.
- [ ] **Interactive Prototype:** Clickable prototype demonstrating navigation from the Home Dashboard into Couple Space, Note Editor, and Knowledge Vault.